

The Burden of Injuries Associated With E-Bikes, Powered Scooters, Hoverboards, and Bicycles in the United States: 2019–2022

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Objectives. To describe the national burden of injuries associated with e-bikes, bicycles, hoverboards, and powered scooters (micromobility devices) in the United States.

Methods. We compared patterns and trends for 1 933 296 estimated injuries associated with micromobility devices from 2019 to 2022 using National Electronic Injury Surveillance System data.

Results. The population-based rates of e-bike and powered scooter injuries increased by 293.0% and 88.0%, respectively. When reported, powered scooter injuries had the highest proportion for alcohol use (9.0%) compared with other modes, whereas e-bike injuries had the highest proportion for motor vehicle involvement (35.4%). Internal injuries were more likely among e-bike diagnoses than hoverboard and bicycle ($P < .05$), but fractures and concussions were more likely among hoverboard diagnoses compared with all other devices ($P < .05$). When helmet use was identified in clinical notes (20.3%), helmet usage was higher among e-bike injuries (43.8%) compared with powered scooter (34.8%) and hoverboard (30.3%) injuries but lower compared with bicycle injuries (48.7%).

Conclusions. The incidence of severe e-bike and powered scooter injuries increased over the 4-year period. Public health stakeholders should focus on improved surveillance and prevention of injuries associated with electric micromobility devices. (*Am J Public Health.* 2024;114(12):1365–1374. <https://doi.org/10.2105/AJPH.2024.307820>)

Micromobility devices, including bicycles, e-bikes, powered scooters, and hoverboards, are gaining popularity as cost-effective, sustainable urban transportation solutions for short-distance travel. While the definition of micromobility is evolving with emerging personal devices and shared systems (e.g., Citi Bike NYC, Lime, Bird), the Federal Highway Administration defines micromobility as “any small, low-speed, human- or electric-powered transportation device.”¹

Micromobility ridership and sales have grown in recent years within the United States.^{2,3} Between 2019 and 2022, e-bike sales increased by 269% and surpassed the sales of electric cars and trucks.³ The number of shared micromobility trips also increased from 64 million trips in 2020 to 112 million in 2021, and, as of 2021, most were by shared electric-powered vehicles (55%, e-scooters; 13%, e-bikes).⁴ This surge in micromobility ridership and availability makes it critical to monitor the burden

of injuries associated with these transportation modes.

The US Consumer Product Safety Commission released a report in September 2023 using National Electronic Injury Surveillance System (NEISS) data, which showed that from 2017 to 2022, emergency department (ED) visits associated with e-bike and powered scooter usage increased linearly, whereas hoverboard-associated visits decreased during this time.⁵ While this report examined several patient and situation

characteristics by micromobility device, the authors did not compare the injury type or risk factors for injuries across devices. Understanding how injury types and risk factors vary by mode can inform ED utilization, resource allocation, and intervention strategies and policies to promote safe micromobility use.

Several other studies used NEISS to determine national estimates of injuries related to micromobility devices.^{6–13} A study by DiMaggio et al. is the only one of these to describe and compare injury patterns and trends across e-bikes, powered scooters, and pedal bicycles, as most studies focus on only 1 micromobility mode or 1 population (e.g., pediatrics).⁷ However, indication of helmet use or alcohol use, 2 risk factors for severe morbidity and mortality in bicycle injury, were not examined.^{14,15} Since the study by DiMaggio et al., NEISS also received a major update that improved the reporting of substance use, powered scooters, hoverboards, and additional patient and situation characteristics.¹⁶

As micromobility availability becomes more widespread in the United States, up-to-date injury surveillance is critical to inform and guide the inconsistent legislation governing micromobility use and to ultimately improve the safety of riders while encouraging usage.¹⁷ This cross-sectional study describes the national burden of injuries associated with micromobility devices by focusing on e-bikes, bicycles, hoverboards, and powered scooters and using 2019–2022 data sourced from NEISS.

METHODS

For this cross-sectional study, we obtained 2019 through 2022 data from the Consumer Product Safety Commission's NEISS query system. NEISS is a

nationally representative stratified probability sample of 96 hospitals in the United States and its territories that contain at least 6 beds and an ED.¹⁸ The stratified sample is based on hospital size and geographic location. Each of the 1 331 870 cases from the 2019–2022 NEISS data are assigned a weight to provide nationally representative estimates of injuries that resulted in an ED discharge. Population estimates were obtained from the US Census Bureau for each year of the study.¹⁹

Data Coding

We generated indicator variables for powered scooter-, e-bike-, and hoverboard-related injuries using product codes and narrative texts (see Appendix Table A [available as a supplement to the online version of this article at <https://ajph.org>] for a list of product codes and search terms). We included sex, race, ethnicity, location of incident, alcohol use, and drug use as reported within the NEISS coding manual.¹⁶ NEISS only collects positive notations for alcohol use and drug use and, thus, negative notations may be included within the “not reported” categories for each variable. We categorized ages into younger than 5, 5 to 9, 10 to 14, 15 to 17, 18 to 44, 45 to 64, 65 to 84, and older than 84 years, with pediatric age groups considered as aged younger than 18 years. These age groups are consistent with available census data to calculate population-based rates.

To generate a motor vehicle variable, we extracted observations for patient narratives that contained “car,” “vehicle,” “truck,” “bus,” or “SUV” terms. For the pedestrian variable, we extracted observations that contained the term “pedestrian.”⁷ We used the disposition

and diagnosis variables as reported within the NEISS coding manual to generate variables for death (dead on arrival, died in the ED, or died after admission), hospitalization (treated and admitted for hospitalization), concussion, internal organ injury, fracture, soft-tissue injury (avulsion, strain, sprain, contusion, abrasion, or laceration), head injury (concussion, internal head injury, external head injury, hematoma, or other head-related injury), and burn (electrical, no specified, scald, chemical, thermal, radiation). A helmet indicator was created using a text-string search approach to extract data from the patient clinical notes, which was operationalized to yes, patient was wearing a helmet; no, patient was not wearing a helmet; or not reported, unknown use of helmet, or no helmet term found within narrative text (code for helmet use and other clinical note-generated variables are available to access in [Zenodo](#)).

Analysis

We used statistical weights to generate population estimates. When the patient had more than 1 micromobility mode reported as involved with the injury, we excluded these unweighted records before analysis ($n = 160$). We used the R package “survey” to generate stratified, weighted, nested, year-adjusted survey estimates.²⁰ Descriptive analyses included total count of injuries that resulted in an ED visit with proportions along with 95% confidence intervals (CIs), which captures the variability based in the NEISS sampling design, stratified by micromobility device. We used survey-adjusted generalized linear models with a log link function (log-binomial regression) to estimate prevalence ratios (PRs) with 95% CIs comparing disposition and diagnoses

between micromobility devices. We used US Census Bureau population estimates to calculate injury and age-specific rates per 100 000 population. We performed all analyses using R statistical software (version 4.3.1; R Core Team 2023, Vienna, Austria). Data and code are available to access in [Zenodo](#).

RESULTS

Out of the 48 857 022 (95% CI = 44 571 027, 53 143 018) total injuries that resulted in an ED visit between 2019 and 2022, there were 1 933 296 (95% CI = 1 717 064, 2 149 529) estimated micromobility injuries. Of these micromobility injuries, most involved bicycles (1 623 143; 95% CI = 1 436 665, 1 809 621), accounting for 33.2 (95% CI = 31.3, 35.2) injuries per 1000 total US ED injuries during the 4 years. Over this 4-year time period, there were an estimated 164 712 (95% CI = 132 797, 196 628) powered scooter injuries accounting for 3.4 (95% CI = 2.7, 4.1) injuries per 1000 total ED injuries. There were also an estimated 56 963 (95% CI = 41 206, 72 721) e-bike injuries or 1.2 (95% CI = 0.9, 1.5) injuries per 1000 total ED injuries, and 88 478 (95% CI = 73 882, 103 073) hoverboard injuries, which accounted for 1.8 (95% CI = 1.6, 2.0) injuries per 1000 total ED injuries between 2019 and 2022.

Demographics

Table 1 reports the demographics of individuals injured across micromobility modes. Of all estimated hoverboard injuries, 76.2% (95% CI = 69.7%, 83.6%) were among those aged younger than 18 years compared with 14.5% (95% CI = 10.3%, 20.7%) of e-bike injuries and 16.1% (95% CI = 12.5%, 20.8%) of powered scooter injuries. By contrast,

57.2% (95% CI = 53.2%, 61.1%) of powered scooter injuries, 48.6% (95% CI = 42.8%, 54.4%) of e-bike injuries, and 30.6% (95% CI = 28.2%, 33.1%) of bicycle injuries were among those aged 18 to 44 years, while 16.7% (95% CI = 14.0%, 19.9%) of hoverboard injuries were among those aged 18 to 44 years (**Figure 1**). Among older adults (65–84 years), the highest proportion of micromobility injuries was bicycle-related (11.8%; 95% CI = 9.7%, 14.3%) and was closely followed by e-bike-related injuries (9.6%; 95% CI = 7.5%, 12.4%). Within the pediatric population (< 18 years), most hoverboard injuries were among those aged 5 to 9 years (41.3%; 95% CI = 38.3%, 44.4%) followed by those aged 10 to 14 years (28.5%; 95% CI = 26.3%, 30.9%).

The proportion of males injured in a powered scooter (63.9%; 95% CI = 60.0%, 67.7%), e-bike (76.9%; 95% CI = 72.9%, 80.5%), and bicycle (73.7%; 95% CI = 72.3%, 75.0%) -related incident was higher compared with females. When patient records included race, the proportions for powered scooter, e-bike, hoverboard, and bicycle injuries were higher among White patients compared with patients of all other reported races, which ranged from 61.2% (95% CI = 53.1%, 68.8%) for powered scooters to 76.0% (95% CI = 72.3%, 79.4%) for bicycles. When ethnicity was reported, the proportion of Hispanic patients was similar for powered scooter (12.7%; 95% CI = 9.9%, 16.1%), e-bike (13.2%; 95% CI = 9.6%, 18.0%), bicycle (11.9%; 95% CI = 9.9%, 14.2%), and hoverboard (12.6%; 95% CI = 9.7%, 16.4%) -associated injuries.

Risk Factors

Most e-bike (80.4%; 95% CI = 73.8%, 85.6%), powered scooter (68.9%; 95%

CI = 61.4%, 75.5%), and bicycle (58.3%; 95% CI = 54.8%, 61.8%) injuries were reported as taking place on a street or highway, as compared with most hoverboard injuries taking place on private property (76.1%; 95% CI = 70.7%, 80.8%). When a positive indication was reported, powered scooter injuries had higher proportions for involving alcohol use (9.0%; 95% CI = 7.2%, 11.1%) and pedestrians (1.1%; 95% CI = 0.8%, 1.6%) compared with e-bike (6.5%; 95% CI = 5.0%, 8.4% and 1.0%; 95% CI = 0.5%, 1.9%), bicycle (4.1%; 95% CI = 3.6%, 4.6% and 0.6%; 95% CI = 0.5%, 0.8%), and hoverboard (0.7%; 95% CI = 0.4%, 1.3% and 0.1%; 95% CI = 0.02%, 0.40%) injuries. E-bike injuries had higher proportions of reported drug use (3.2%; 95% CI = 2.3%, 4.4%) and motor vehicle involvement (35.4%; 95% CI = 29.3%, 41.9%) compared with powered scooter (1.8%; 95% CI = 1.4%, 2.4% and 25.4%; 95% CI = 19.6%, 32.1%) and bicycle (1.7%; 95% CI = 1.4%, 2.0% and 21.9%; 95% CI = 20.1%, 23.9%) injuries. When helmet use was identified in clinical notes (20.3%), indication of helmet use was higher among e-bike injuries (43.8%; 95% CI = 37.5%, 50.3%) compared with powered scooter (34.8%; 95% CI = 30.7%, 39.0%) and hoverboard (30.3%; 95% CI = 19.0%, 44.7%) injuries but lower compared with bicycle injuries (48.7%; 95% CI = 45.3%, 52.1%).

Disposition and Diagnoses

In descriptive analyses (**Table 2**), 12.7% (95% CI = 9.4%, 17.0%) of e-bike-related injuries resulted in hospitalization compared with 10.2% (95% CI = 9.0%, 11.6%) for powered scooters, 10.5% (95% CI = 9.6%, 12.1%) for bicycles, and 3.2% (95% CI = 2.5%, 4.0%)

TABLE 1— Descriptive Characteristics for Micromobility Injuries in the National Electronic Injury Surveillance System: United States, 2019–2022

Characteristic	Bicycle (n = 1 623 143), % (95% CI)	E-Bike (n = 56 963), % (95% CI)	Hoverboard (n = 88 477), % (95% CI)	Powered Scooter (n = 164 712), % (95% CI)
Age, y				
<5	3.4 (3.0, 3.8)	0.7 (0.2, 1.8)	4.3 (3.5, 5.4)	0.8 (0.5, 1.3)
5–9	11.1 (9.9, 12.4)	2.4 (1.6, 3.8)	41.3 (38.3, 44.4)	4.0 (3.0, 5.3)
10–14	14.5 (13.2, 15.9)	6.9 (5.2, 9.1)	28.5 (26.3, 30.9)	7.4 (5.8, 9.4)
15–17	5.0 (4.6, 5.4)	4.5 (3.3, 6.1)	2.1 (1.6, 2.9)	3.9 (3.2, 4.8)
18–44	30.6 (28.2, 33.1)	48.6 (42.8, 54.4)	16.7 (14.0, 19.9)	57.2 (53.2, 61.1)
45–64	23.1 (22.1, 24.2)	27.3 (24.0, 30.8)	5.8 (4.5, 7.5)	19.4 (17.8, 21.2)
65–84	11.8 (9.7, 14.3)	9.6 (7.5, 12.4)	1.1 (0.7, 1.8)	6.2 (4.9, 7.7)
>84	0.5 (0.4, 0.7)	0.06 (0.0, 0.2)	0.0 (<0.01, <0.01)	1.1 (0.7, 1.7)
Not reported	0.04 (0.0, 0.1)	0.0 (<0.01, <0.01)	0.0 (<0.01, <0.01)	0.0 (<0.01, 0.1)
Sex				
Male	73.7 (72.3, 75.0)	76.9 (72.9, 80.5)	44.7 (42.3, 47.1)	63.9 (60.0, 67.6)
Female	26.3 (25.0, 27.6)	23.1 (19.5, 27.1)	55.3 (52.9, 57.7)	36.1 (32.4, 40.0)
Intersex/nonbinary	0.009 (<0.01, 0.03)	0.0 (<0.01, <0.01)	0.0 (<0.01, 0.1)	0.1 (<0.01, 0.3)
Not reported	0.0 (<0.01, <0.01)	0.0 (<0.01, <0.01)	0.0 (<0.01, <0.01)	0.0 (<0.01, <0.01)
Race				
American Indian/Alaska Native	0.7 (0.5, 0.9)	0.4 (0.1, 1.2)	0.6 (0.2, 1.5)	0.6 (0.4, 1.0)
Asian	2.4 (2.0, 3.0)	2.9 (1.8, 4.8)	1.6 (0.9, 2.6)	3.0 (2.1, 4.4)
Black/African American	17.2 (13.7, 21.3)	30.9 (19.4, 45.4)	20.1 (16.6, 24.2)	31.7 (23.3, 41.6)
Native Hawaiian/Pacific Islander	0.2 (0.1, 0.3)	0.1 (0.01, 0.6)	0.4 (0.1, 1.1)	0.2 (0.1, 0.5)
Other	3.5 (2.6, 4.5)	3.0 (1.4, 6.5)	4.3 (2.9, 6.4)	3.1 (2.0, 4.9)
White	76.0 (72.3, 79.4)	62.7 (50.6, 73.4)	73.1 (68.9, 77.0)	61.2 (53.1, 68.8)
Not reported	34.2 (26.3, 43.1)	25.6 (16.7, 37.2)	37.8 (27.6, 49.3)	19.2 (14.5, 25.1)
Ethnicity				
Hispanic	11.9 (9.9, 14.2)	13.2 (9.6, 18.0)	12.6 (9.7, 16.4)	12.7 (9.9, 16.1)
Not Hispanic	88.1 (85.8, 90.1)	86.8 (82.0, 90.4)	87.4 (83.6, 90.3)	87.3 (83.9, 90.1)
Not reported	38.2 (29.6, 47.6)	52.7 (37.4, 67.5)	38.9 (28.6, 50.3)	34.4 (23.0, 47.9)
Location of injury				
Place of recreation or sport	12.8 (10.8, 15.2)	3.6 (2.0, 6.5)	5.1 (3.1, 8.2)	2.4 (1.6, 3.6)
Private property	13.2 (11.4, 15.2)	4.1 (2.4, 6.9)	76.1 (70.7, 80.8)	8.5 (6.3, 11.3)
Public property	15.1 (12.6, 18.0)	11.9 (8.5, 16.3)	6.7 (4.5, 9.9)	20.1 (15.6, 25.5)
School or daycare	0.6 (0.5, 0.7)	0.1 (<0.01, 0.4)	0.2 (0.1, 1.1)	0.2 (0.1, 0.7)
Street or highway	58.3 (54.8, 61.8)	80.4 (73.8, 85.6)	11.9 (7.99, 17.2)	68.9 (61.4, 75.5)
Not reported	41.5 (37.1, 46.0)	31.0 (24.5, 38.3)	50.1 (42.9, 57.2)	36.3 (30.4, 42.6)
Alcohol use				
Yes	4.1 (3.6, 4.6)	6.5 (5.0, 8.4)	0.7 (0.4, 1.3)	9.0 (7.2, 11.1)
Not reported	95.9 (95.4, 96.4)	93.5 (91.6, 95.0)	99.3 (98.7, 99.6)	91.0 (88.9, 92.8)
Drug use				
Yes	1.7 (1.4, 2.0)	3.2 (2.3, 4.4)	0.1 (<0.01, 0.5)	1.8 (1.4, 2.4)
Not reported	98.3 (98.0, 98.6)	96.8 (95.6, 97.7)	99.9 (99.5, 100.0)	98.2 (97.6, 98.6)
Motor vehicle involved				
Yes	21.9 (20.1, 23.9)	35.4 (29.3, 41.9)	3.5 (2.7, 4.7)	25.4 (19.6, 32.1)
Not reported	78.1 (76.1, 79.9)	64.6 (58.1, 70.7)	96.5 (95.3, 97.4)	74.6 (67.9, 80.4)

Continued

TABLE 1— Continued

Characteristic	Bicycle (n = 1 623 143), % (95% CI)	E-Bike (n = 56 963), % (95% CI)	Hoverboard (n = 88 477), % (95% CI)	Powered Scooter (n = 164 712), % (95% CI)
Pedestrian involved				
Yes	0.6 (0.5, 0.8)	1.0 (0.5, 1.9)	0.1 (0.02, 0.40)	1.1 (0.8, 1.6)
Not reported	99.4 (99.2, 99.5)	99.0 (98.1, 99.5)	99.9 (99.6, 100.0)	98.9 (98.4, 99.2)
Helmet use				
Yes	48.7 (45.3, 52.1)	43.8 (37.5, 50.3)	30.3 (19.0, 44.7)	34.8 (30.7, 39.0)
No	51.3 (47.9, 54.7)	56.2 (49.7, 62.5)	69.7 (55.3, 81.0)	65.2 (61.0, 69.3)
Not reported	80.5 (77.9, 82.8)	68.9 (62.7, 74.5)	97.1 (95.9, 97.9)	78.3 (72.9, 82.9)

Note. CI = confidence interval.

for hoverboards. However, analytic analyses revealed that e-bike-related ED visits were only significantly more likely to be severe enough to result in hospitalization (PR = 4.0; 95% CI = 2.8, 5.8) and diagnoses of a soft tissue injury (PR = 1.4; 95% CI = 1.3, 1.5) and head injury (PR = 1.3; 95% CI = 1.1, 1.7) compared with hoverboard-related ED visits. In addition, internal injuries were more likely to be diagnosed among e-bike ED visits compared with hoverboard (PR = 1.7; 95% CI = 1.2, 2.2) and bicycle ED visits (PR = 1.2, 95% CI = 1.01, 1.5). E-bike ED visits were also 5 times more likely to result in a burn diagnosis compared with bicycle ED visits (PR = 5.0; 95% CI = 1.8, 14.3). Lastly, hoverboard ED visits were more likely to result in diagnoses of fractures and concussions compared with e-bike (PR = 1.4; 95% CI = 1.2, 1.6 and PR = 2.8; 95% CI = 1.6, 5.0), powered scooter (PR = 1.3; 95% CI = 1.2, 1.4 and PR = 1.5; 95% CI = 1.0, 2.05), and bicycle ED visits (PR = 1.4; 95% CI = 1.3, 1.5 and PR = 1.3; 95% CI = 1.05, 1.7).

Trends

The population-based rate of e-bike-related injuries increased between 2019 and 2022 by 293.0% (Appendix Figure A, Appendix Table B).

This linear relationship was consistent across all age groups (Figure 2), except for among adults aged older than 84 years, for whom rates were vanishingly small and insignificant (Appendix Table C). While the population-based rate of powered scooter injuries increased by 88.0% over the 4-year period, by 2022, the rate of injuries began to plateau (Appendix Figure A, Appendix Table B), with some evidence of a decrease in injuries among the group aged 18 to 44 years in late 2022 (Figure 2, Appendix Table C). The population-based rates of hoverboard and bicycle injuries were highest before 2021 but since then have declined, with hoverboard injury rates decreasing by 26.2% from 2019 to 2022 (Appendix Figure A, Appendix Table B).

DISCUSSION

In this study, we described and compared the burden of injuries associated with e-bikes, bicycles, hoverboards, and powered scooters within the United States during a time of significant advancement for the electric micromobility industry. Overall, the rate of hoverboard-related injuries decreased over the 4 years and among the pediatric population who experienced the

greatest burden of these injuries. This may be explained by hoverboards becoming less popular or could be evidence to support pediatric injury-prevention efforts such as the statement released in 2018 by the American Academic of Pediatrics warning of the dangers of hoverboard use.²¹ Conversely, the population-based rate of e-bike-related injuries increased by nearly 300% between 2019 and 2022, which closely parallels the increases in e-bike sales and shared e-bike ridership.^{3,4} The rate of powered scooter-associated injuries almost doubled, which may be explained by the increase in shared e-scooter trips across the United States.⁴ The increase in electric micromobility injuries might also be attributable to the lack of access, education, and regulation for protective equipment as shared micromobility systems are not required to provide helmets to users. The legislation for where to ride these devices or for riding these devices under the influence is also lacking and inconsistent,¹⁷ and this could lead to greater risks to users such as riding in closer proximity to motor vehicles or while intoxicated.

Those seeking ED care for e-bike, powered scooter, and bicycle injuries

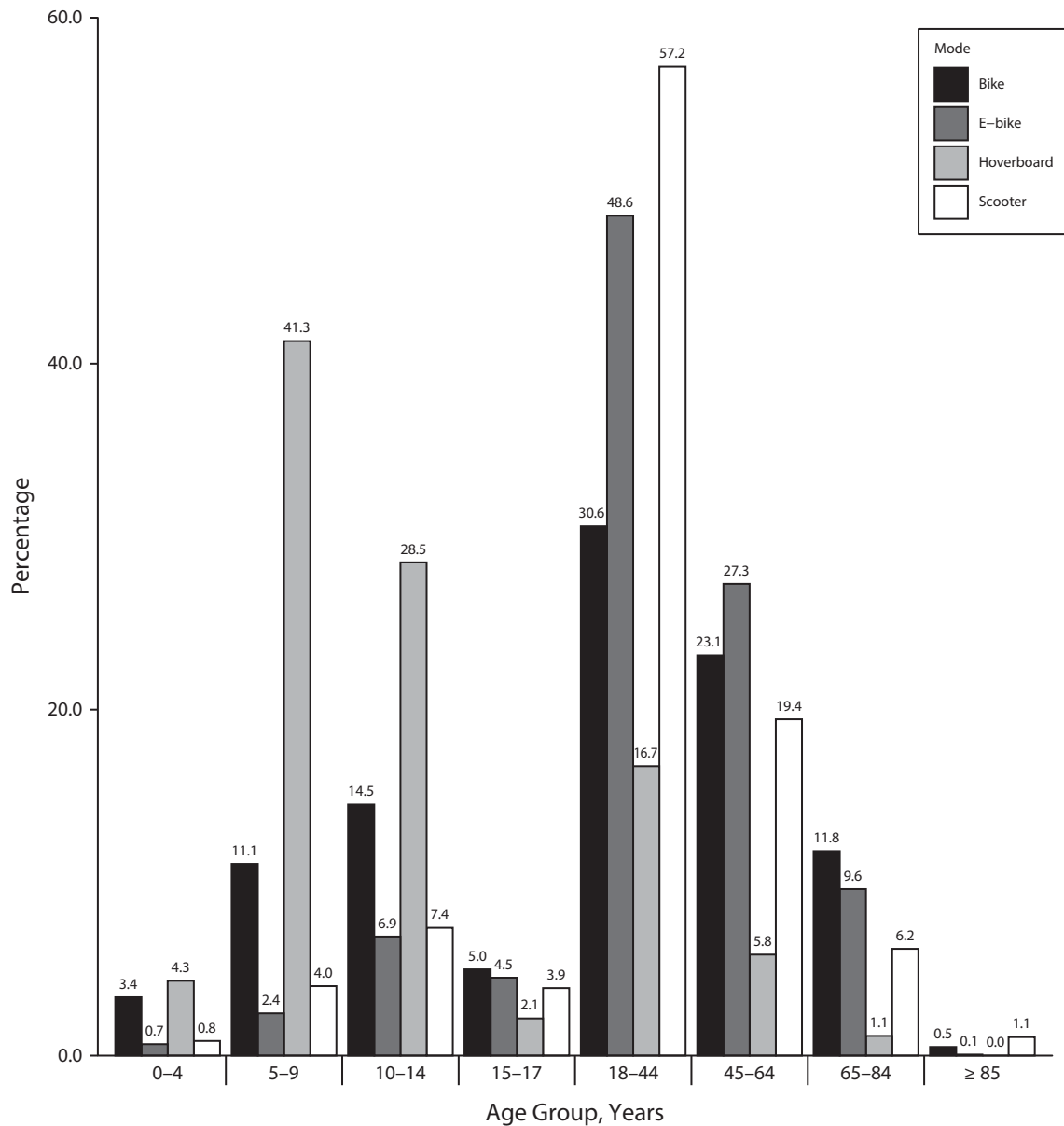


FIGURE 1— Percentage of Bicycle, E-Bike, Hoverboard, and Powered Scooter Injuries by Age: National Electronic Injury Surveillance System, United States, 2019–2022

were more often middle-aged, male, and White. One explanation for these findings is that men tend to use e-bikes and e-scooters more than women and engage in riskier behaviors such as alcohol use, which may place them at greater risk of injury.^{5,22} Men have also been found to travel faster on e-scooters and to be more likely to

use alcohol, drugs, or a smartphone while riding e-scooters compared with women.²³

In alignment with the Consumer Product Safety Commission's recent report, we found that while the highest proportion of injuries across all micro-mobility modes was among White patients compared with patients of

other races, the proportion of e-bike and powered scooter injuries among Black/African American patients was disproportionately high compared with the percentage of Black Americans in the general population (about 13%).^{5,24} However these results should be interpreted with caution because of the incomplete race and ethnicity data, a

TABLE 2— Descriptive Characteristics of Disposition and Diagnoses for Micromobility Injuries in the National Electronic Injury Surveillance System: United States, 2019–2022

Characteristic	Bicycle (n = 1 623 143), % (95% CI)	E-Bike (n = 56 963), % (95% CI)	Hoverboard (n = 88 477), % (95% CI)	Powered Scooter (n = 164 712), % (95% CI)
Disposition				
Died	0.15 (0.1, 0.21)	0.03 (<0.01, 0.2)	0.00 (<0.01, <0.01)	0.14 (0.1, 0.4)
Hospitalized	10.5 (9.6, 12.1)	12.7 (9.4, 17.0)	3.2 (2.5, 4.0)	10.2 (9.0, 11.6)
Diagnoses				
Concussion	3.1 (2.7, 3.4)	1.5 (0.9, 2.5)	4.1 (3.2, 5.3)	2.8 (2.2, 3.6)
Internal	14.0 (13.0, 15.1)	17.4 (13.6, 22.1)	10.5 (8.9, 12.3)	13.9 (12.4, 15.6)
Fractures	29.2 (28.1, 30.5)	29.7 (25.9, 33.8)	41.6 (39.2, 44.1)	31.8 (29.9, 33.7)
Soft tissue	51.5 (50.4, 52.6)	54.7 (51.1, 58.3)	39.8 (37.3, 42.3)	52.9 (50.5, 55.3)
Burns	0.06 (0.0, 0.1)	0.3 (0.1, 0.8)	0.0 (<0.01, 0.1)	0.13 (0.1, 0.3)
Head injury	18.8 (17.8, 19.7)	21.4 (17.6, 25.7)	15.9 (14.2, 17.7)	19.8 (18.0, 21.6)

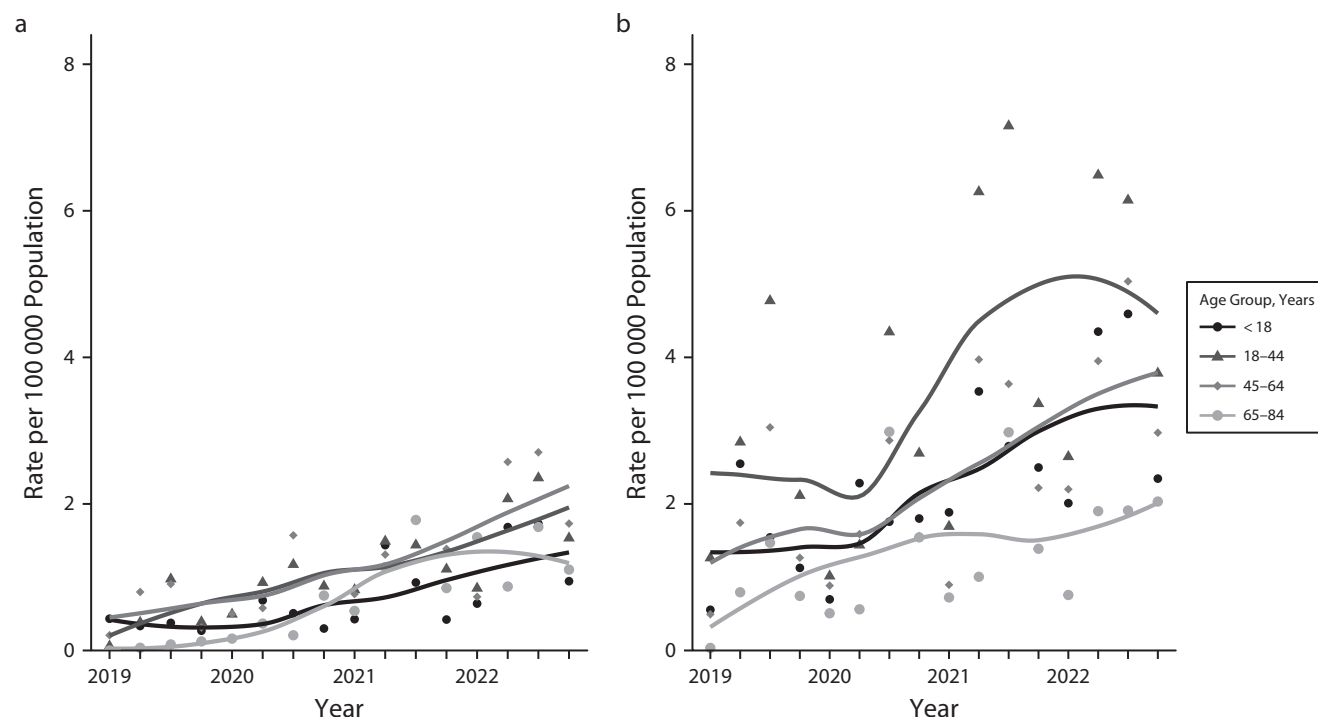
Note. CI = confidence interval.

well-known limitation of NEISS and other injury surveillance systems.²⁵ Given this limitation, smaller studies with access to complete race and ethnicity data could seek to identify vulnerable populations who may rely on certain

micromobility devices. In New York City, for instance, approximately 65 000 commercial delivery workers, often referred to as “deliveristas,” primarily utilize e-bikes, scooters, and mopeds to complete their deliveries, but their

safety and occupational risks have rarely been studied.^{26,27}

We also observed that powered scooter-associated injuries exhibited the highest prevalence of alcohol use, followed by e-bike-associated injuries.

**FIGURE 2—** Quarterly Rate per 100 000 US Population by Age for (a) E-Bike and (b) Powered Scooter Injuries: National Electronic Injury Surveillance System, 2019–2022

These findings suggest that alcohol involvement may be a public health concern in electric micromobility-related injuries. Yet, again, these results should be interpreted with caution as NEISS only collects data on positive indications of alcohol use and codes a subjective measure of acute alcohol use based on if clinicians' notes mention alcohol involvement. Thus, this existing method for collecting alcohol use likely results in selection bias and measurement error.²⁸ To improve our understanding of the risk of alcohol use in electric micromobility injuries, future studies are needed that incorporate objective measures of acute alcohol use such as blood alcohol content and more robust epidemiological study designs to determine if alcohol use is a risk factor for these injuries. Micromobility injuries that involve alcohol may be in close proximity to alcohol-serving establishments, nightlife districts, or university or college campuses where micromobility shared systems are typically placed.^{4,29} One population-level strategy to prevent injuries associated with these devices may be to focus on improving active transportation infrastructure (e.g., traffic slowing, protected bike lanes, wider sidewalks, docking stations) near high-usage locations such as downtown areas and nightlife districts where alcohol consumption is promoted.

We found that when helmet use was reported, in about 20% of patient narratives, the proportions for bicycle and e-bike injuries involving helmet usage was higher than the proportions for injuries associated with powered scooters or hoverboards. The lowest proportion of helmet use was observed among hoverboard injuries, and these injuries were also more likely to be diagnosed as concussions compared

with other modes. These data suggest that concussions in hoverboard injuries remain a particular concern for the pediatric population for whom hoverboard injuries were most prevalent in our study. Policy attention to mandating protective gear use among children riding hoverboards may be needed especially if these devices remain popular. The finding for helmet usage among e-bike injuries was somewhat surprising as bicycle shared systems typically do not provide users with helmets, which is likely for hygiene, cost, and logistical reasons. Given the large percentage of missing helmet-use data, the estimates in our study are likely an inaccurate representation of helmet usage across injuries associated with these devices. Yet, our findings for helmet usage among e-bike injuries was identical to that of a recent report,¹³ which reviewed all patient narratives for indication of helmet use in the NEISS data set between 2017 and 2022. In addition, a recent study examining the utility and reliability of a large language model and the text-string search approach we used for extracting helmet status from clinical notes showed that the text-string search approach had high validity when compared with a human-coded gold standard.³⁰ Complete and accurate data for characterizing helmet usage and other risk factors among micromobility users is critical for informing targeted intervention strategies and policies. As mandatory helmet laws are controversial and likely difficult to pass,³¹ public health experts could focus attention on helmet use interventions or understanding the safety effects of active transportation infrastructure to prevent micromobility injuries.

In this study, we also found that internal injuries were more likely to be diagnosed in e-bike-related ED visits compared with visits associated with hoverboards and bicycles. In addition, burns were more likely to be diagnosed in e-bike ED visits compared with bicycle ED visits. DiMaggio et al. similarly found that e-bike injuries had higher proportions for internal injuries compared with bicyclist and powered scooter injuries, but soft tissue injury proportions were much lower among e-bike ED visits than results in our study.⁷ Differences in our findings may ultimately be attributable to the rise in e-bike injuries in recent years. However, evidence is growing to suggest that e-bike injuries are more severe and injure more body parts compared with bicycle injuries,³² which may reflect the increased speed of travel and weight of e-bikes compared with other micromobility modes.³³ While we found very few burn diagnoses within our study, the highest proportion estimates were for burns were among e-bike-associated ED visits, and there were few burn diagnoses for e-bike injuries overall, which was surprising given the media attention to deadly and injurious fires from lithium-ion batteries used by these devices.³⁴

Limitations

This study added to the very little existing knowledge base on important sociodemographic and risk factor variables that might be contributing to micromobility vehicle-related injuries. This gap in the literature is likely because existing national public-use data sets such as NEISS have only recently begun to collect data on electric micromobility devices. While this study improved our limited understanding of the burden of injuries for users of

micromobility devices, there are limitations to be noted. First, several estimates for variables across micromobility modes were small (e.g., death, burns) and may be unstable or unreliable. This is an inherent limitation of the NEISS sampling method and, thus, nationally representative studies that do not rely on a probability sample.¹⁸ Ideally, data that provide information on exposure before injury would allow for stronger epidemiological study designs that inform on the risks associated with these micromobility modes. This may ultimately require building partnerships with rideshare systems, emerging crowdsourced data, or collecting retrospective data from injured users of these devices.

As previously discussed, measurement error is also a limitation within our study, particularly for variables that were generated exclusively through narrative text (e.g., motor vehicle- or pedestrian-involved, helmet use). Natural language processing or large language models applied to clinical notes could supplement the validated text string approach we developed for helmet status coding to improve its efficiency.^{28,35} Overall, while emerging micromobility devices have the potential to provide population-level health, climate, and transportation benefits, implementing uniform definitions of micromobility devices and improving the collection of complete risk factor data within injury surveillance is urgently needed to accurately inform micromobility policies and interventions.

Public Health Implications

In summary, this study highlights numerous public health concerns associated with the burgeoning micromobility industry. While less prevalent than

traditional bicycle injuries, the rate of e-bike injuries increased fourfold over the 4-year period, and the rate of powered scooter injuries nearly doubled. Given the ongoing growth of the electric micromobility sector and recent historic investments in active transportation by the federal government, it is imperative for the public health community to advance surveillance and prevention of micromobility-related injuries. *AJPH*

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K. G. Burford acquired the data, conducted data analyses, and led the writing of the article. All authors conceptualized the study, interpreted

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CONFLICTS OF INTEREST

The authors have no conflicts of interest to disclose.

HUMAN PARTICIPANT PROTECTION

The study was exempt from institutional review board approval because this study used publicly available, de-identified data.

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